SPE-229320-MS

Role of AI-Driven Reservoir Characterization in Critical Operational Decision-Making: An Onshore UAE Case Study

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Technology has become an essential part of the oilfield industry, with Oil Operator companies focusing on the deployment of innovative initiatives and leveraging the benefits of their implementation. The demand for automation has increased significantly in both the Upstream and Downstream segments of the industry. Artificial Intelligence (AI)-driven processes are particularly relevant in this context. This work highlights the application of AI in characterizing and understanding oil reservoir rock properties. Traditionally, the integrated analysis of well log data is done manually and can take days to complete, but a fully automated machine-learning-driven approach can perform this task in a much shorter time. In the Petrophysics industry, Well-Bore Image (WBI) logs require extensive analysis due to their detailed geological content, which makes them more time-consuming to interpret compared to other logs. Ensuring the timely availability of high-quality outputs is crucial for the efficient design and execution of subsequent well operations, as demonstrated in the case study presented in this paper. AI-generated results are both time-efficient and detailed, providing sharp and clear outputs for operational needs.

Comprehensive log data acquisition, including Well Bore Imaging (WBI), was performed in candidate horizontal lateral wells situated in heterogeneous carbonates of Lower Cretaceous age. The captured information-spectrum encompassed triple-combo, sonic, and nuclear magnetic resonance logs, as well as drilling, mud-log, and core data from the pilot offset and horizontal producer well-bores. Machine learning was utilized to execute WBI-based textural heterogeneity analysis. AI-driven key deliverables included quantifying diagenetic porosities and extracting rock permeability indices. These results were subsequently integrated with other down-hole information to develop a thorough understanding of dynamic flow outcomes. This comprehensive approach was crucial feed for designing an effective Lower Completion strategy within the required timeframe.

AI-driven analysis has reduced the time required for tasks while maintaining quality. What used to take days or weeks can now be completed in hours. Cost optimization has also been achieved as a result of reducing dependency on contractors. The results have provided better insights into production flow profiles. AI-based outputs showed stronger correlation with flow logs compared to conventional rock

property outputs, enhancing understanding of reservoir behaviour. Utilizing AI-derived analytical results in operational decisions has become the best practice in our organization. One of the practical applications includes aiding in the design of Lower Completion for newly drilled wells, where actual or test data has not yet been obtained. It is important to note that AI machine learning is an iterative process, expected to improve over time with more testing and feedback.

The refined workflow utilizes supervised machine learning and computer vision to analyse electrical well-bore images at each depth, enhancing texture analysis and image porosity evaluation for permeability prediction. Through finite element Darcy experiments, the system simulates permeability variations along the wellbore under different boundary conditions. This AI-driven approach assesses the impact of various pore structures on overall permeability, applying machine learning algorithms to classify and quantify these effects accurately. This methodology provides depth-specific permeability predictions, supporting exploration and management in carbonate reservoirs.

Introduction

This paper examines the application of Artificial Intelligence (AI) in reservoir characterization. Deep learning methods were applied to Well Bore Image (WBI) logs from two geological formations. The study utilized an AI/ML application developed by ADNOC Technical Center and AIQ. The AI application conducted carbonate textural heterogeneity analysis. The paper presents the outcomes of using machine learning techniques and describes their impact on operational decision-making processes.

Datasets & Downhole Geology

Primary data (WBI logs) were gathered from two wells for this exercise. Additional datasets—triple-combo logs, petrophysical results, drilling data, flow and mud logs—from the same wells were integrated into the analysis. Core data from nearby pilot vertical wells were also used. See Table-1 for dataset details and Fig. 1 for well locations.

Table 1—Summary of Datasets used in this study.

Well Details	*Location*	*Well Deviation*	*Datasets Used*	*Summarized Geology*	*Data Quality*
Well-1 (Oil Producer)	Onshore Abu Dhabi	Horizontal	BHI, Triple-combo, Drilling, Flow logs & offset well Core data	Lower Cretaceous heterogenous carbonates	Good
Well-2 (Oil Producer)	Onshore Abu Dhabi	Horizontal	BHI, Triple-combo, Drilling, Mud logs & offset well Core data	Lower Cretaceous tight carbonates	Good

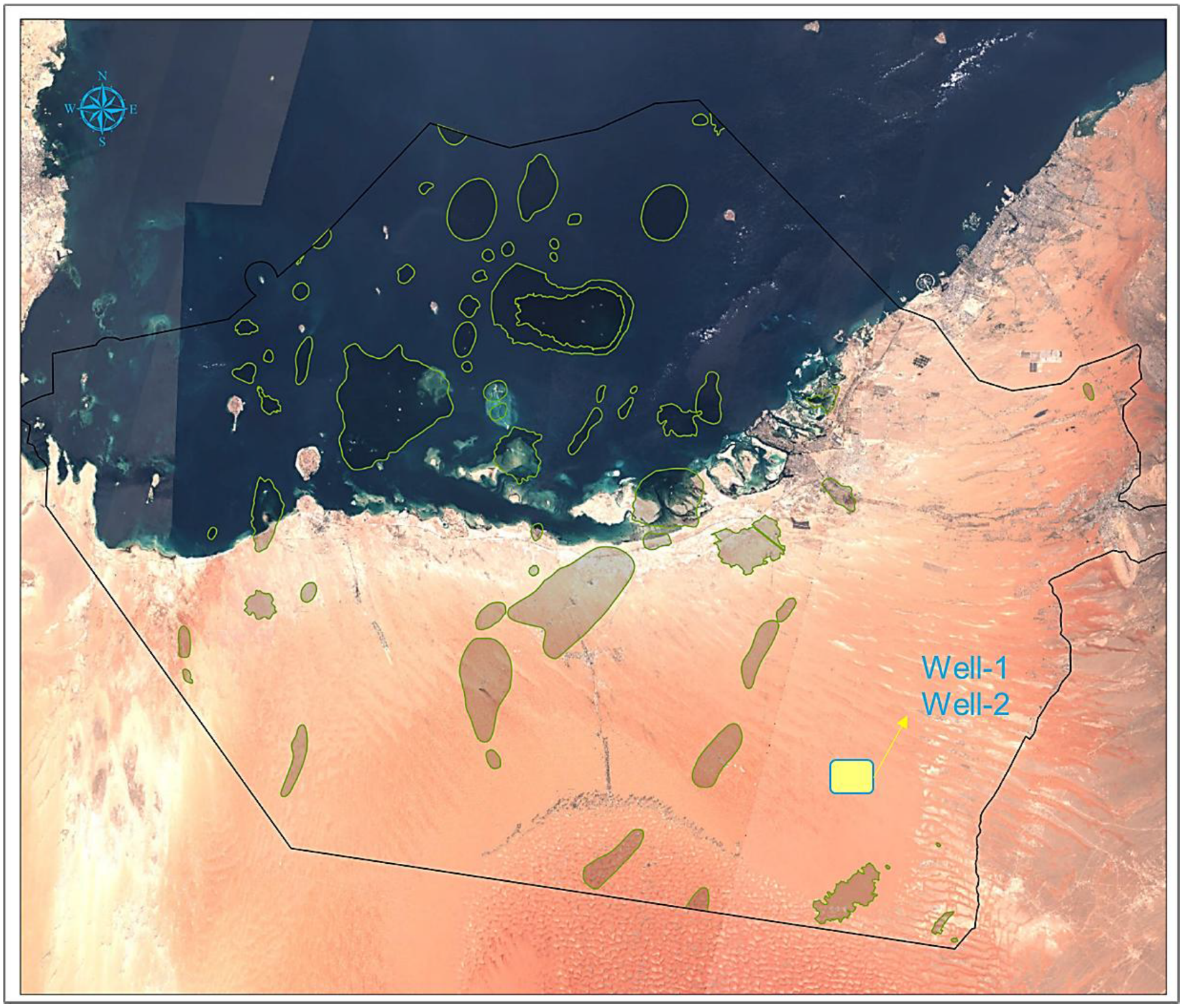

Figure 1—UAE map showing location of candidate wells.

Work Synopsis

AI-based rock permeability analysis was conducted in Wells 1 & 2. In both instances, the outputs from deep learning were combined with associated down-hole data to reach an integrated conclusion that would go on to feed into critical operational decision-making.

Challenge Statement

Characterizing carbonate reservoirs remains one of the primary challenges within the Oil & Gas sector. Well Bore Imaging (also referred to as Bore Hole Imaging (BHI) in this paper) is an essential technology for gaining a comprehensive understanding of carbonate rock properties (Newberry, WM et al, 1996). Traditional WBI log interpretation involves meticulous, depth-by-depth analysis of structural, sedimentary, and diagenetic features, a process that demands considerable attention and can extend over several days or even weeks depending on the length of the logged interval.

Furthermore, current algorithms designed to analyse carbonate heterogeneity—such as those used to estimate secondary porosity and permeability index from image log data—often provide only indirect solutions and can be time-intensive. While automated dip-picking methods have long existed, their results frequently include excessive noise, limiting their effectiveness.

As WBI-derived outputs play a crucial role in informing key operational decisions, there is a clear need for timely delivery of quality-controlled interpreted data. Such demands have driven the development and adoption of AI-based methodologies to improve efficiency and reliability in reservoir characterization. How Machine Learning (ML) BHI-based rock property outputs provided critical feed into optimized designing of Lower Completions (Smart Liner including) – comprises the core of this paper.

Methodology

Well Bore Image (WBI) Logs – a Background

Borehole imaging derived from micro-resistivity logs plays a critical role in evaluating in-situ stress conditions and geological heterogeneity in the vicinity of wells. Traditional logging tools, typically offering resolutions greater than 1 foot, can overlook smaller features such as vugs and anomalies, especially when lacking complete perimeter coverage. In contrast, micro electrical imaging provides significantly higher resolution (5.08 mm) alongside full azimuthal coverage, making it well-suited for detailed detection of formation heterogeneity (Fig. 2). These images accurately record both well depth and orientation; variations in pixel intensity correspond to differences in local porosity and permeability. For instance, in water-based mud environments, areas of high-intensity pixels denote denser rock phases, whereas lower intensities indicate more porous zones.

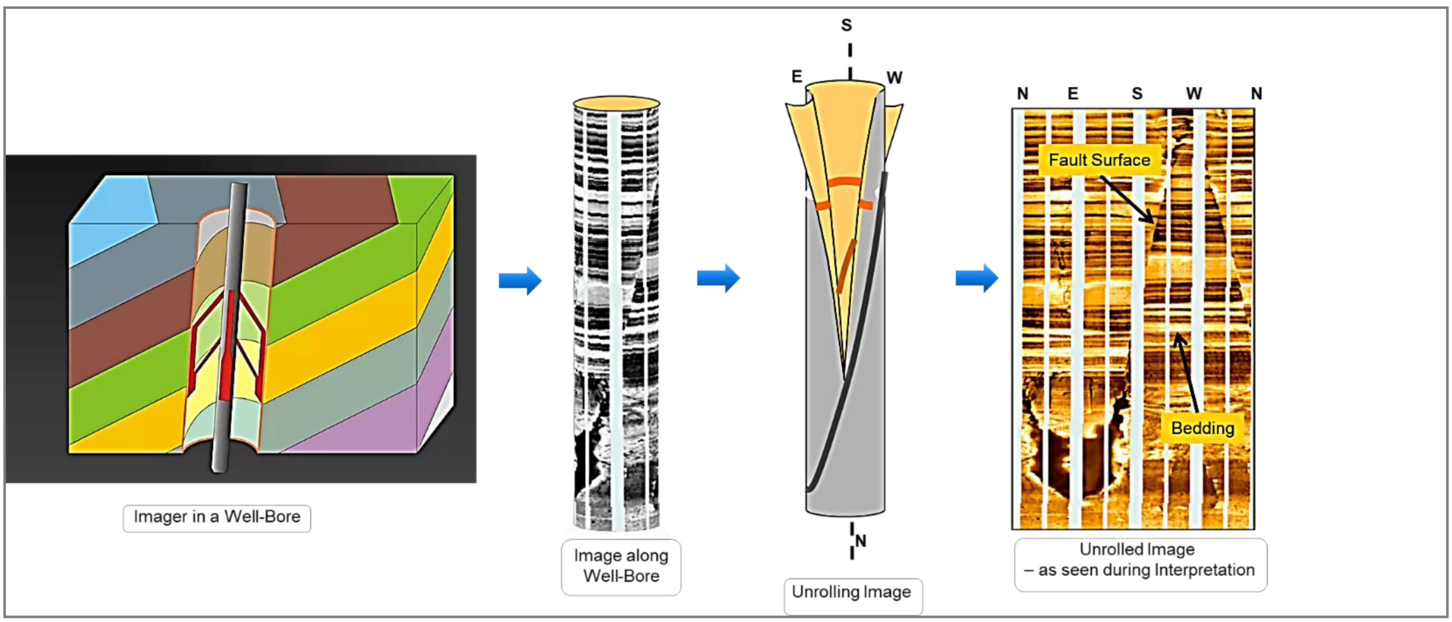

Figure 2—Display & Orientation of a WBI log (modified after Vickerman, K., 2016)

Micro-resistivity borehole images provide geoscientists with detailed subsurface insights, surpassing the limitations of conventional logs. They reveal macro-scale features from tectonic events (fractures, faults), diagenesis (vugs, stylolites), and artificial events (drilling-induced fractures, breakouts) (Sharland, P.R et al, 2001 & Al Sharhan, A.S et al. 1997, 2005).

Geoscientists can identify important features for hydrocarbon recovery, but doing so manually is slow and repetitive.

AI-driven Bore Hole Image Analysis

Machine learning has been applied to address challenges in characterizing carbonate reservoirs. Deep learning enables accurate segmentation of geological features from large, labelled borehole image datasets without relying on fixed intensity thresholds. This reduces manual analysis and can improve reliability in borehole image interpretation, potentially transforming geoscience workflows (Petrov, A. et al, 2023).

A recent supervised deep learning method uses a fully convolutional network where experts delineate feature curves for model training. However, this approach struggles with bi-dimensional features like vugs and breakouts, overlapping elements such as fractures, and cases requiring both thickness and profile characterization (Petrov, A. et al, 2023).

Semantic Segmentation for Borehole Heterogeneity Extraction. Semantic segmentation enables the classification of image pixels into predefined categories to facilitate advanced analysis. Convolutional neural networks (CNNs) generate probability maps that represent the likelihood of each pixel being associated with a particular class. In borehole image analysis, each category corresponds to distinct heterogeneities, and this approach can be adapted to accommodate various types of borehole heterogeneity.

The CNN is trained using binary masks, where background pixels are labelled as 0 and pixels of interest as 1. The training process seeks to optimize a loss function that minimizes discrepancies between the binary mask and the output probability map from the CNN. Binary masks for different classes are consistently created following established expert annotation protocols. Upon completion of training, the CNN module infers the most probable class membership for each individual pixel. Class-specific thresholds are established to interpret the results: pixels exceeding the threshold are assigned to the relevant class, while those below the threshold are designated as background. This process clearly delineates regions within borehole images that correspond to varying heterogeneities, as illustrated in Fig. 3.

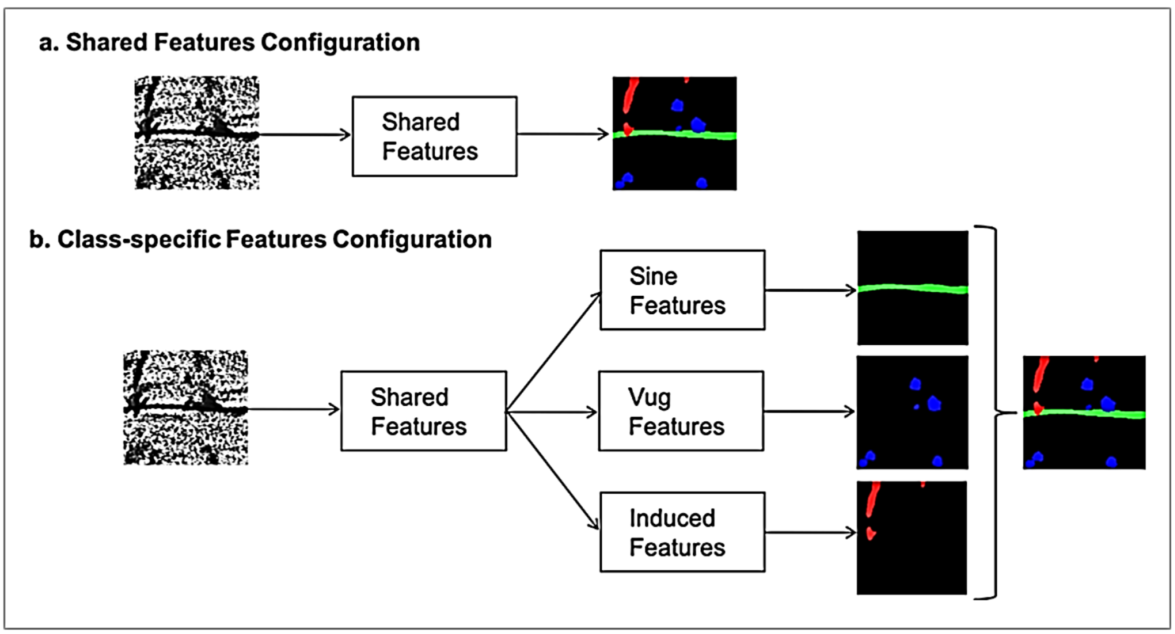

Figure 3—Description of the CNN Semantic Segmentation module in two different configurations: a, multi-task shared features configuration and b, single-task class specific configuration (Noufal, A., 2022).

Employing semantic segmentation with CNNs and binary masks streamlines the automated classification and identification of diverse heterogeneities found in borehole images. This methodology delivers valuable insights for geoscientists, enhancing both the efficiency and accuracy of subsurface formation analysis during hydrocarbon exploration. Two configurations of the CNN model have been identified: one utilizes shared features to produce an all-classes mask, while another employs separate branches to yield class-specific features. The former configuration is more straightforward to train and is typically adopted initially, whereas the latter, although requiring more extensive training, improves classification accuracy for challenging, closely related classes (Petrov, A. et al., 2023).

The process of determining the connectivity of geological features involves a three-step methodology using binary masks derived from semantic segmentation. Initially, the union (U(m)) and intersection (I(m)) of class-specific masks are calculated to identify overlapping or coinciding areas of different geological features within the borehole image. Subsequently, connected regions within the union (U(m)) are extracted to encompass each intersection region identified in (I(m)). This step aims to group pixels belonging to the same geological feature, even if they appear in multiple classes. Finally, these connected regions (Fig. 4) within the union (U(m)) are used to extract individual connected features from the initial binary masks, thereby effectively isolating and characterizing each geological heterogeneity (Petrov, A. et al, 2023).

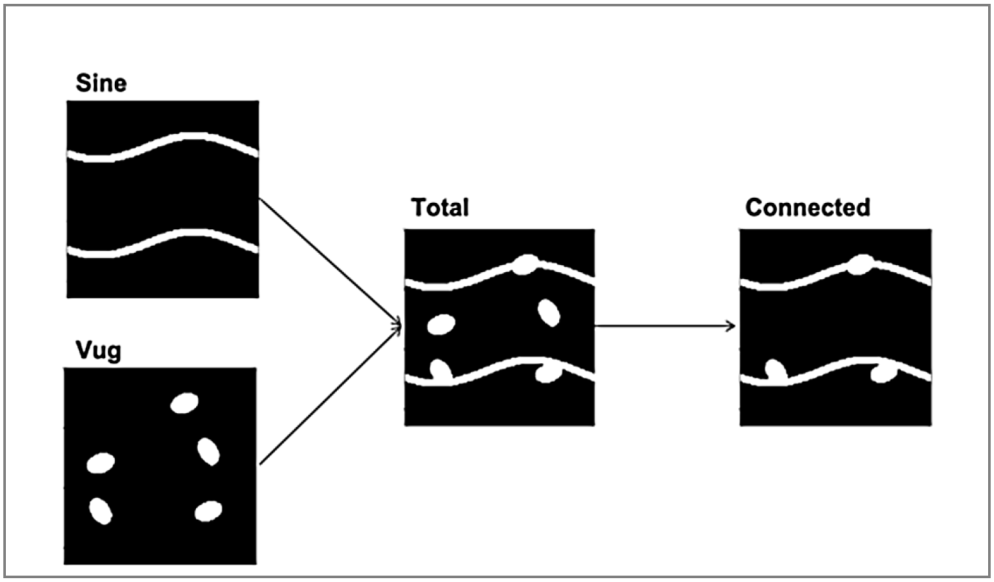

Figure 4—Description of Connectivity Module: intersections between sines and vugs are used to extract connected regions from the total image U(m) (Noufal, A., 2022).

Borehole Image derived porosity. The wellbore static image normalizes pixel values between 0 and 1 to assess the respective contributions of the matrix and heterogeneities to overall porosity. Primary porosity is governed by the matrix, while secondary porosity consists of heterogeneous regions identified using semantic segmentation techniques. For each row, primary porosity is calculated by aggregating the matrix pixel values and dividing by the total number of pixels in that row. In a similar manner, secondary porosity is determined by summing the heterogeneity pixel values and normalizing this sum by the row's total pixel count (Noufal, A. et al, 2022).

The findings of our analysis are presented in a porosity versus depth plot, as shown on the right side of Fig. 5. To clarify the comparative contributions of primary and secondary porosity, the secondary porosity is visually superimposed onto the primary porosity. Owing to substantial fluctuations in porosity between adjacent rows, a moving average method has been employed to display the data, utilizing a window span of one foot for this purpose. This window span is adaptable and can be modified according to the observed heterogeneity scale within the borehole images. The approach facilitates a more precise representation and understanding of porosity variations along the borehole depth, offering valuable insights into reservoir porosity distribution and the influence of heterogeneities (Noufal, A. et al, 2022).

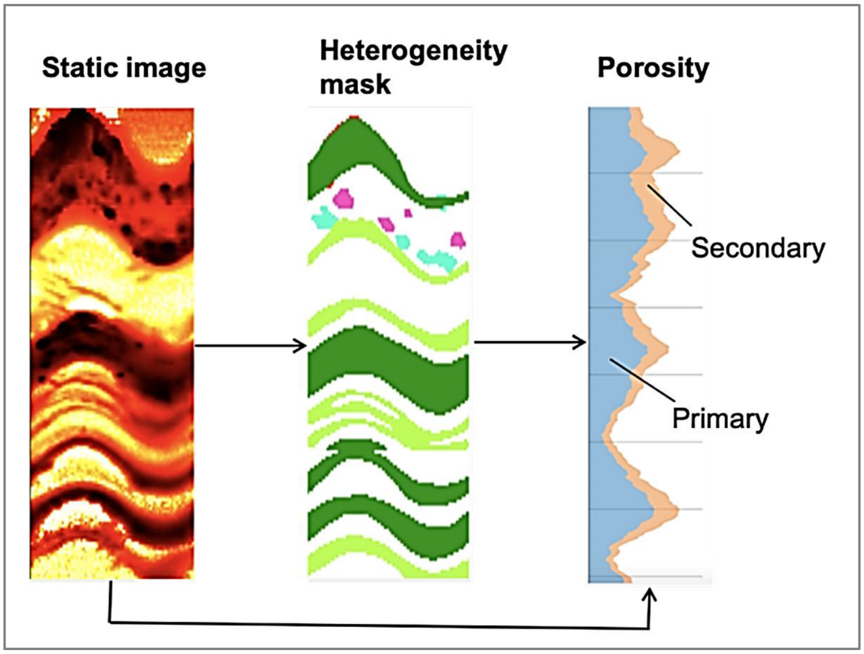

Figure 5—Primary and secondary porosity calculation from the static image and the heterogeneity mask computed though semantic segmentation (Noufal, A., 2022).

Permeability estimation from borehole images. The described algorithm employs the Finite Element Method (FEM) to address the diffusion equation on a matrix derived from the static WBI. The workflow consists of four steps: (i) segmenting to identify natural and induced features within the image; (ii) removing these features from the static image; (iii) using an inpainting algorithm to fill the imputed zones; and (iv) applying Darcy's law to determine the permeability of the matrix and features (Noufal, A. et al, 2022). Further details are provided in Fig. 6.

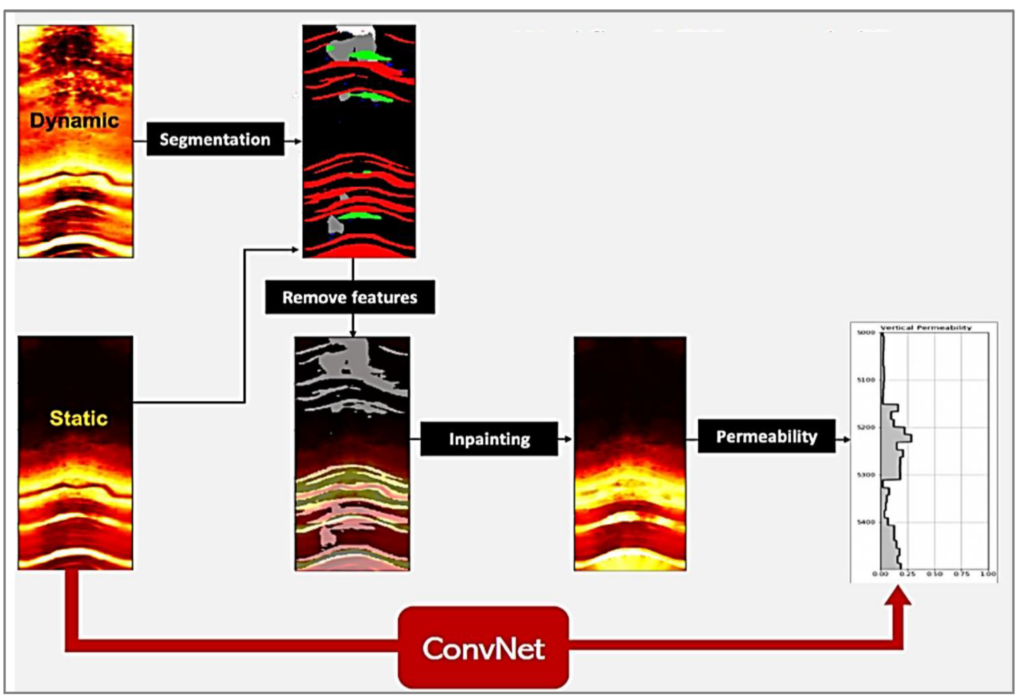

Figure 6—Permeability prediction using ConvNet (Petrov, A., 2023).

Data Analysis & Results Discussion

Presented below are two case examples that illustrate the use of machine learning in integrated petrophysical analysis.

Case 1: Horizontal Oil Producer in Lower Cretaceous heterogeneous Carbonates. The 3000 ft long oil-producer well-bore had high-resolution resistivity image acquired on Logging-While-Drilling (LWD) mode along with triple-combo logs. The well was analysed for structural features (including natural fractures, faults and layering).

Detailed interpretation revealed a structurally 'quiet' well-path with few or no natural fractures. Reservoir structural dip computed from bedding surfaces is gentle (< 5 degrees).

The horizontal well was placed in the optimum part of the target reservoir sub-unit, with near-consistent log porosity values. At places where the well-bore got close to the dense unit beneath the reservoir section, subtle variations in log responses were observed.

The AI-based permeability index was estimated along the well-bore length. A conventional permeability log was also calculated.

The standard log permeability depicted a near-uniform response almost mimicking the porosity log response- throughout the entire horizontal well-length. The AI-derived permeability index, however, predicted the first half of the well-bore to be property-wise better than the second half (Fig.7).

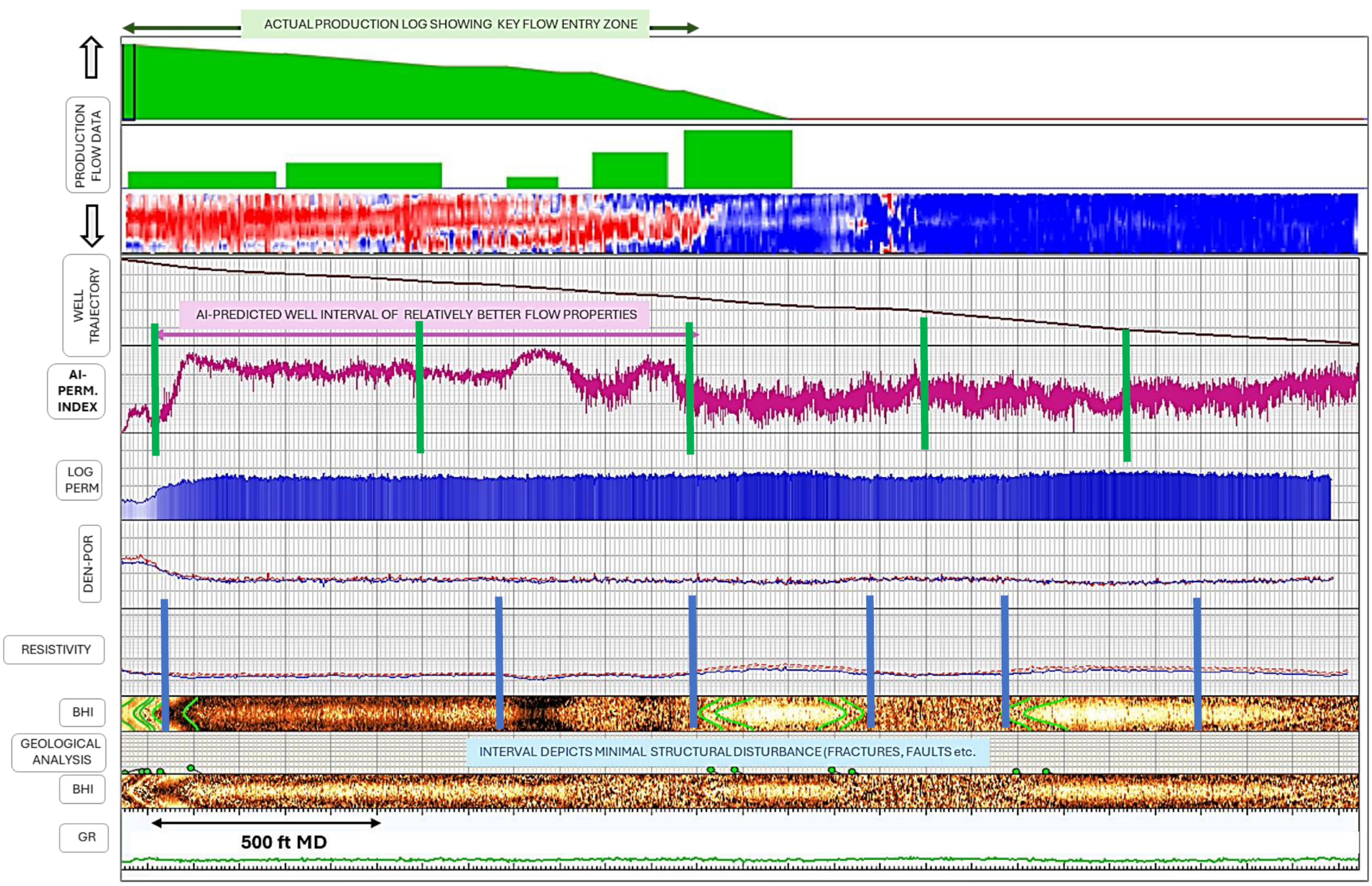

Figure 7—Case 1: Horizontal oil producer in Lower Cretaceous heterogeneous Carbonates: Integrated analysis plot.

Later, the well was tested, and a production log was run to understand the flow profile and zones of oil and/or water contribution.

A plot depicting all the data acquired in the horizontal well-section including the AI-derived Permeability Index from WBI, conventional log permeability, and PLT flow zones – is shown (Fig.7).

The actual zonal flow contribution showed positive correlation with AI-derived rock flow property prediction. A holistic analysis incorporating all the mentioned datasets also revealed that the flow appeared to be matrix-dominated (with little or no influence of natural fractures). The afore-mentioned results were then critical input feed into the Lower Completion design strategy (Fig. 8).

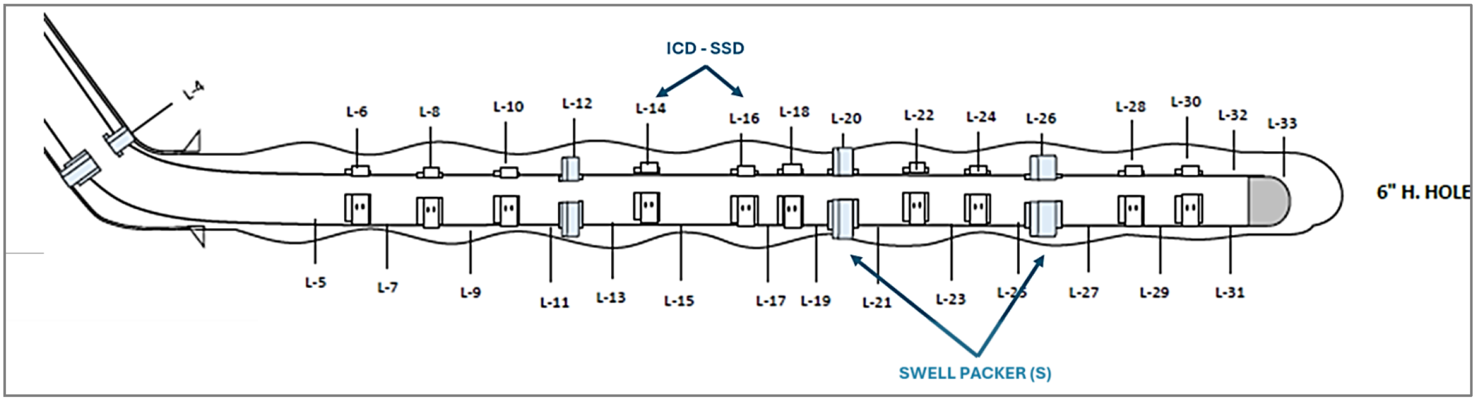

Figure 8—Case 1: Lower Completion (LC) Design arrived at, based on AI-results-driven Integrated Petrophysical Analysis.

In summary, utilizing only log permeability and conventional BHI outputs for LC design would result in segmentations as depicted by the "blue" bars in Fig. 7. Incorporating the AI-BHI Permeability Index, which aligns with MPLT results, enables optimized segmentations as illustrated by the "green" bars. This approach reduces the number of swell packers required and allows for precise ICD opening design in each segment to ensure effective water production control. Consequently, the AI Permeability Index has been instrumental in achieving accurate and permanent segmentation of LC within the well.

Case 2: Horizontal Oil Producer in Lower Cretaceous tight Carbonates. The 4000 ft long oil-producer well-bore had high-resolution resistivity image acquired on Logging-While-Drilling (LWD) mode along with triple-combo logs. The well was analysed for structural features (including natural fractures, faults and layering).

Detailed interpretation revealed the presence of at least 22 minor faults and associated natural (planar) fractures (Fig.9). The faults encountered have caused juxtaposing of facies resulting in abrupt variations in density-porosity log signatures. Fracture fill nature identified range from electrically conductive (open) to electrically resistive (cemented). The faults and planar (tectonic) fractures intercepted have steep dips (60 – 90 degrees). Reservoir structural dip computed from bedding surfaces reach unusually high values for the region (up to 20 degrees). This variation in dip magnitude along well-path appears to be clearly related to presence of nearby lineaments.

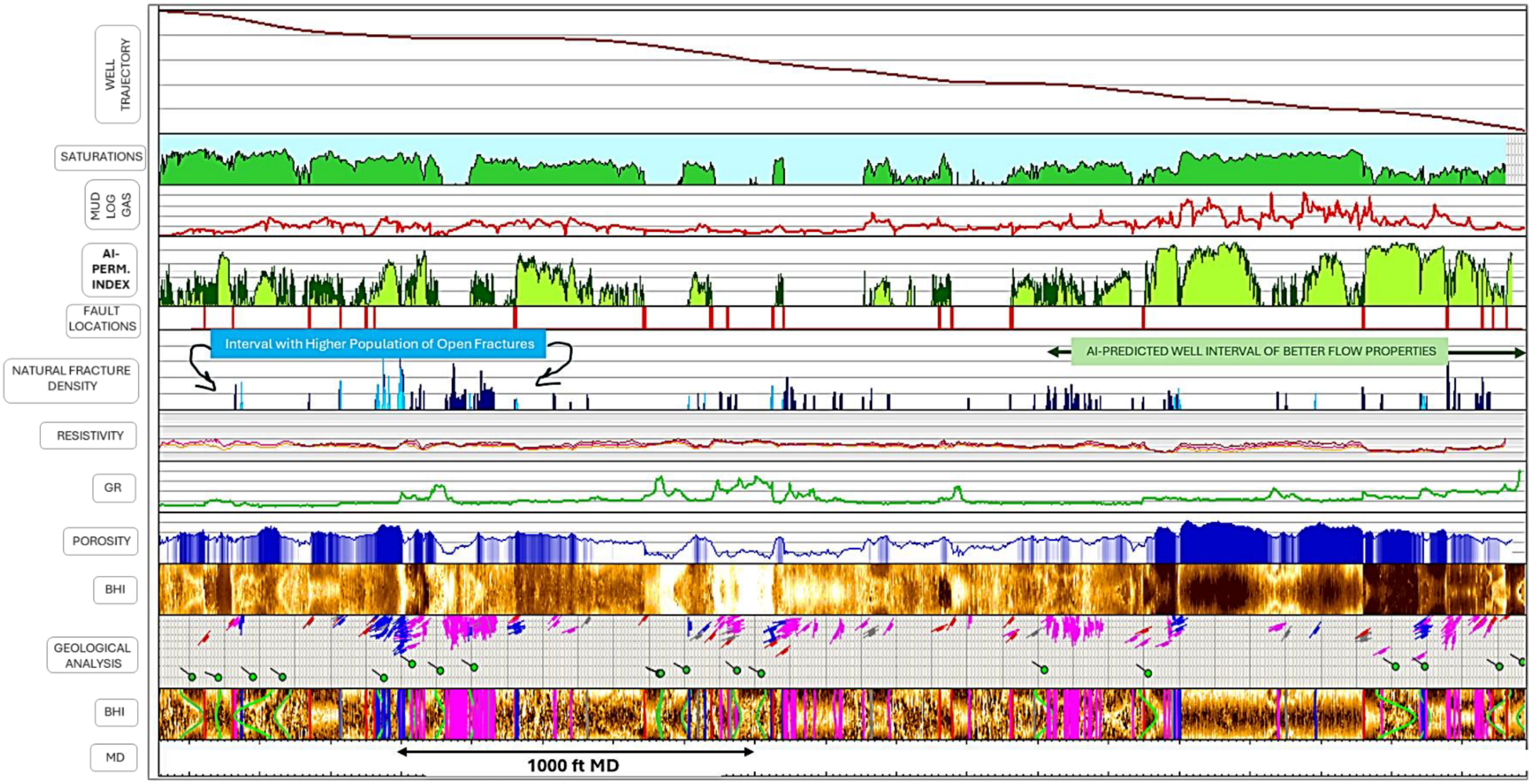

Figure 9—Case 2: Horizontal oil producer in Lower Cretaceous tight Carbonates: Integrated analysis plot.

To better comprehend reservoir matrix properties, the AI-based permeability index was estimated along the well-bore length. The AI-derived permeability index predicts better flow properties towards the TD section of the well bore (Fig.9).

Integrated analysis also does not rule out contribution of natural fractures and minor faults to production – especially - near the upper interval close to the heel section of the well (despite this region being relatively inferior in rock properties compared to the lower section of the well-bore).

To achieve uniform stimulation across an extended horizontal well section, a Smart Liner lower completion design was implemented. Prior to drilling, a preliminary completion design was developed based on predicted reservoir properties and prepared at the wellsite. Following drilling, the final Smart Liner design was refined using actual subsurface data.

To prevent over-stimulation near the heel and ensure effective stimulation propagation toward the toe, reservoir porosity and permeability data were analysed, along with fracture and fault interpretations derived from borehole imaging. AI-predicted permeability data were also incorporated to enhance the precision of the final design.

As a result, zones with poor reservoir quality and high water saturation were isolated using blank pipes. The placement and spacing of perforated liners were optimized based on porosity distribution and AI-driven permeability predictions. In high-permeability and fracture-dense zones, perforations were sparsely distributed to avoid acid concentration, whereas in low-permeability and toe regions, perforations were densely arranged to maximize stimulation. This approach enabled a uniform stimulation effect throughout the entire horizontal section (Fig.10).

Completion Design	
Completion Type	Single Oil Producer
Tubing Design	4.5″ x 3.5″
Open Hole	6″ Open Hole, +/- 3900ft
Lower Completion Type	7″ Liner Hanger 4.5″ Smart Liner and Swell Packers
Liner Joint	30.5′ Joint Length Blank Liner(33) and 3mm Perforated Liner(86)

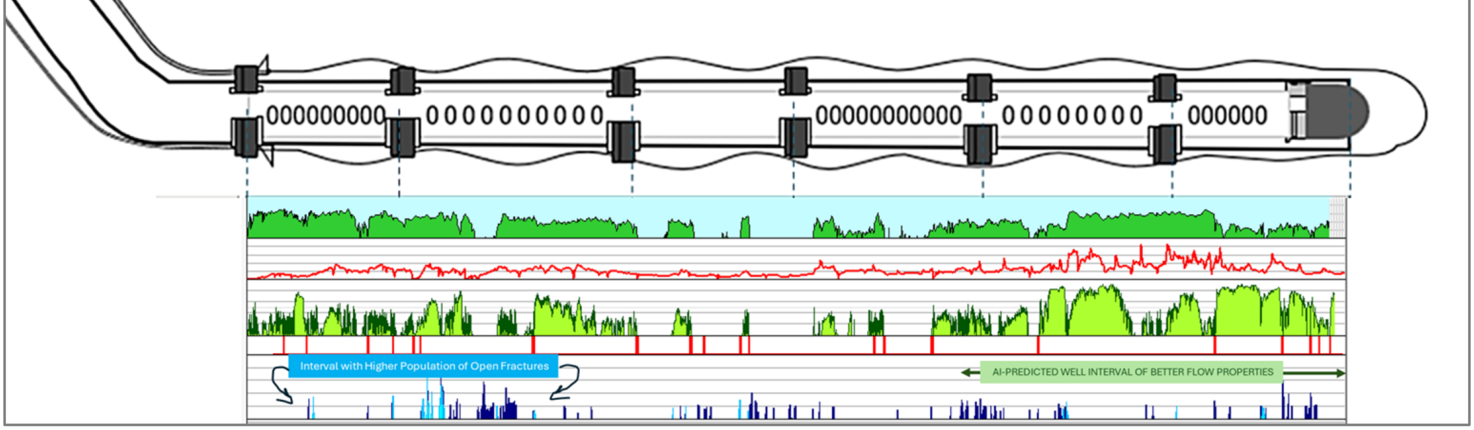

Figure 10—Case 2: Smart Liner Design Strategy based on AI-driven integrated analysis.

Conclusion & Recommendation

The application of deep learning techniques in reservoir characterization has resulted in significant cost and time savings, enhanced process efficiency, and enabled more rapid yet well-informed operational decision-making. Insights derived from image log-based integrated analyses have been effectively incorporated into crucial operational phases such as lower completion design planning and the prediction of production and injection flow profiles.

Traditional analysis and interpretation platforms tend to be labour-intensive or require indirect, time-consuming workarounds. In contrast, the AI-driven approach has substantially reduced the timelines for WBI-based downhole geological interpretation *from days or weeks to just minutes or a few hours.* This has allowed for meaningful cost reductions through the elimination of contractor work orders, resulting in annual CAPEX savings *of tens of thousands of dollars.* Furthermore, the adoption of AI-driven integrated downhole geological analysis into subsequent well operation design strategies is generating *production gains valued at millions of dollars.*

Improved AI-derived outputs like BHI continuous permeability need to be a crucial input into Field Development studies for reservoir engineering, as it affects the estimation, simulation, and optimization of the reservoir performance and behaviour. For example, as seen from our case studies, BHI derived permeability log is of a much higher resolution (0.2 inch) than conventional log permeability (0.5 ft) and contains a plethora of information on reservoir rock character. While core-derived permeability represents ground-truth, it suffers from sample bias. Log-derived permeability is affected by lower resolution leading to missing out on various carbonate textural and structural detail. This is where deep-learning-guided image log-based permeability outputs find its significance.

Like any other Artificial Intelligence venture, this methodology also has room for growth. For example, quantifying the BHI permeability index, through proper calibration, to output in Darcy units – is work in progress at the time of writing this paper. For instance, dip picking' by the AI-App requires further maturing in certain reservoir rock types. Deep learning as a continuous (iterative) process is ongoing for several geological Formations. With time, volume of iterations and resultant 'learning', the AI App is expected to better its deliverables.

Acknowledgement

We wish to express our deepest gratitude to Aalya Ahmed Mohammed Al Jasmi, our GIS expert, for her valuable support with location map preparation. Again, this endeavour would not have been possible without the continued encouragement and blessings of ADP / AYP Senior Management, for which we are immensely indebted to.

We also take this opportunity to give credits to the AI-App inventors (Dr. Abdelwahab Noufal and Khalid Obaid) and AIQ team who generated this solution. Just as importantly, we thank ADNOC for allowing us to publish this work.

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